

Saam Haghighat-Grami

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📍 Roswell, GA

🔧 Interests – Autonomous, Mobile, Collaborative Robotics

🎓 Graduating December 2026

🌐 LinkedIn – saam-haghighat-grami

🐙 GitHub – Saam-Grami

🌐 Portfolio – saam-grami.github.io

🔒 Active Military Secret Security Clearance

Education

- **Kennesaw State University** January 2022 – May 2022, January 2024 – Present
Mechatronics Engineering, B.S. – Graduation December 2026 CGPA: 4.00
- **Marine Corps Communication Electronics School** August 2022 – October 2023
Ground Electronics Transmission Systems Maintainer CGPA: 4.00

Technical Skills

Autonomy & Robotics: ROS2, Gazebo, Nav2, ArduPilot, PX4, Mission Planner, UAV/UGV integration, RTAB Map, UR5e, Control Theory

Embedded & Hardware: ESP32, Arduino, Raspberry Pi 5, OpenMV, LiDAR, RGB-D cameras, IMUs, ad hoc networking, Nema 23 stepper, N20 motors, MentorPi, Tello

Languages & Tools: C++, Python, MATLAB, LaTeX, Linux/WSL2, Git, VS Code, Arduino IDE, SolidWorks, Onshape

Perception & ML: PyTorch, TensorFlow, YOLO, CNNs, LSTMs, OpenCV, ORB-SLAM3, sensor fusion

Manipulators & Automation: Denavit-Hartenberg parameters, forward and inverse kinematics, April tags, trajectory planning, Ladder Logic, TIA Portal, PLCs, HMI development

🌐 **Portfolio** — saam-grami.github.io — Full publication list, in-progress projects (multi-drone swarm sim, GPS-denied UAV-UGV localization), and additional technical detail.

Research & Engineering Experience

- **MEAP – Mass Estimation Autonomous Pathfinding** November 2025 - Present
Robotics Software Architect & Project Coordinator – AgriSys Lab, Dr. Muhammad Hassan Tanveer
 - Architected a UAV-UGV collaborative search-and-rescue system, fusing aerial map stitching with SLAM-based ground navigation to estimate obstacle mass and volume in real time.
 - Led a 5-person team building the UAV autonomy stack, validated on a 2 m² area and scaling to 25 m²; leading the mass-estimation ML pipeline, targeting 85% mission success on obstacles up to 0.5 kg.
- **DARB – Drone Arm Robotic Backpack (Personal Project)** May 2025 - Present
Sole Designer and Builder
 - Designed a backpack-mounted robotic arm that autonomously captures, stows, and relaunches a drone mid-flight, targeting a 30-second capture-to-relaunch cycle.
 - Built and flight-tested the full system with created custom airframe using ArduPilot on a Matek h743 WLITE through MAVLink. Attached OpenMV orients drone to robotic arm. Drone and Arm communicate autonomously through ESP-NOW. – across repeated live test flights.
- **Trajectory Forecasting for Floating Debris in Waterways** June 2026 - August 2026
Funded Undergraduate Researcher – Summer Undergraduate Research Program, Dr. Matt Marshall
 - Built a real-time vision system that detects, tracks, and forecasts floating river debris trajectories from riverbank video to direct a collection platform toward interception.
 - Developed a CNN + LSTM forecaster combining optical-flow surface velocimetry with YOLO26 detections, producing calibrated multi-step trajectory forecasts with track re-identification.
- **SoilBus – Soil Monitoring Probe Network** August 2025 - December 2025
Lead Software Developer – AgriSys Lab, Dr. Muhammad Hassan Tanveer
 - Designed an AD HOC soil-monitoring mesh network with auto-discovering probes, letting any single node offload the entire network's data.
 - Achieved full mesh connectivity in 15s and ~5s per-node data offload; validated BFS-based routing on 5 physical nodes, scaling to 26 simulated.

Publications

Author on 3 works in autonomous systems research – 1 published (IEEE SoutheastCon 2026, IEEE Xplore), 2 first-author manuscripts in preparation for peer reviewed computer vision venue and IROS 2027. *Full citations and DOIs at portfolio above.*

Industry & Military Experience

- **EOSYS Industrial and Manufacturing System Integrator** May 2025 – August 2025
COOP Marietta GA
 - Built an HMI display generator that produced 40+ P&ID screens for plant operators.
 - Reviewed and modified PLC programs across two production shops plus the PLC trainer system.
- **United States Marine Corps Reserve** August 2022 - Present
Ground Radio Repair Specialist Marietta GA
 - Troubleshoot, maintain, and operate USMC radio systems in time-sensitive conditions; active Secret clearance.
 - Lead a squad of 8 Marines in the 4th Combat Readiness Regiment, responsible for training, accountability, and readiness.